Machine learning-based noninvasive estimation of cardiac output from peripheral blood pressure waveforms: *in vivo* validation

Vincent Morier-Genoud

Supervisor(s): Prof. Dr. Raphael Sznitman, Dr. Lydia Aslanidou
Institution(s): University of Bern, ARTORG Center for Biomedical Engineering Research
EPFL, the Swiss Federal Institute of Technology in Lausanne,
Laboratory of Hemodynamics and Cardiovascular Technology
Examiners: Prof. Dr. Raphael Sznitman, Dr. Lydia Aslanidou

Introduction

The cardiac output (CO) represents the total blood flow pumped by the heart. In the ICU, haemodynamic monitoring and fluid therapy guidance critically rely on accurate CO assessment. The gold standard is thermodilution with a Swan-Ganz catheter, but its invasiveness has motivated the search for less invasive alternatives. Estimating CO from arterial blood pressure (ABP) waveforms using convolutional neural networks (CNNs) trained on *in silico* data has been increasingly investigated, with promising results. However, the lack of high-quality *in vivo* datasets has so far limited their validation, motivating the present work.

Materials and Methods

Thermodilution Cardiac output (TCO) measurements were temporally matched to ABP segments from the MIMIC-III database and enriched with additional patient data. An automated signal processing algorithm, combining quality criteria analysis and morphological classification, was developed to identify a representative wave from each segment. The resulting *in vivo* dataset was used to evaluate CNN models pre-trained on *in silico* data. New CNNs were then trained on the collected *in vivo* dataset, and a stepwise feature addition analysis quantified the impact of each input by tracking changes in MAE, with features introduced sequentially as follows: ABP waveform, its first and second derivatives, heart rate, mean blood pressure, age, and gender.

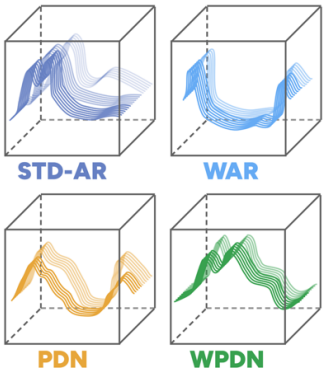

Fig. 1 ABP waveform morphological subspaces derived using single value decomposition (SVD).

Results

A dataset of 5,275 samples from 579 patients was assembled, containing only morphologically standard and quality-validated ABP waves, linked to TCO target values and additional patient data. A CNN trained directly on this *in vivo* dataset achieved an MAE of 0.95 L/min and a correlation of r = 0.67.

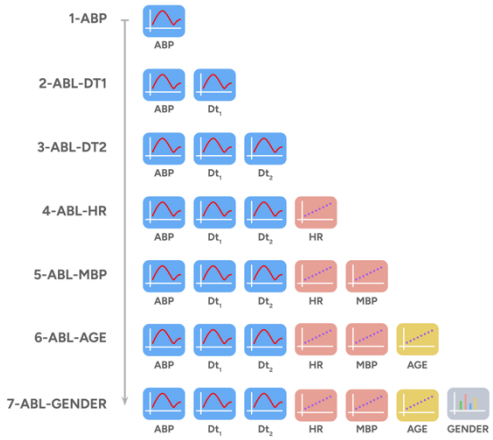

Fig. 2 Progressive feature addition for CNN models (ABP, dt_1, dt_2, HR, MBP, age, gender).

Discussion

The quality of the dataset and its generation algorithm was evaluate both through visual inspection of the samples and by verifying CO prediction using mathematical estimators against a reference dataset from MIMIC-II. The evaluation of pre-trained *in silico* models confirmed that their promising results on synthetic data did not generalise as well as expected *in vivo*. The stepwise feature addition analysis highlighted the importance of inputs such as ABP signal derivatives, as well as patient age and gender.

References

Aslanidou, L., Rovas, G., Mohammadi, R. et al. Machine learning-enabled estimation of cardiac output from peripheral waveforms is independent of blood pressure measurement location in an *in silico* population. Sci Rep 15, 25562 (2025).